

1 Ceptifi Suspen 5% INJECTABLE 2 COMPOSITION Ceftiofur.....50 mg Exp.qsto.....1 ml DESCRIPTION: The finished product is a brownish, oily suspension. PHARMACODYNAMICS Ceftiofur is a third generation cephalosporin, which is active against many Gram-positive and Gram-negative bacteria, including <i>β</i>-lactamases producing strains. Ceftiofur acts by inhibiting the bacterial cell wall synthesis, thereby exerting bactericidal properties. Cell wall synthesis is dependent on enzymes that are called penicillin-binding proteins (PBPs). Bacteria develop resistance to cephalosporins by four basic mechanisms: 1) altering or acquiring penicillin binding proteins insensitive to an otherwise effective <i>β</i>-lactam; 2) altering the permeability of the cell to <i>β</i>-lactams; 3) producing <i>β</i>-lactamases that cleave the <i>β</i>-lactam ring of the molecule and destroy the drug. Some <i>β</i>-lactamases, documented in Gram-negative enteric organisms, may alter ceftiofur MICs to varying degrees to third and fourth generation cephalosporins, including ceftiofur. In addition to <i>β</i>-lactam inhibitor combinations, and first and second generation cephalosporins, ceftiofur is active against the following microorganisms which are involved in respiratory diseases in swine: <i>Pasteurella multocida</i>, <i>Actinobacillus pleuropneumoniae</i> and <i>Streptococcus suis</i>. <i>Bordetella bronchiseptica</i> is intrinsically non-susceptible to ceftiofur. It is also active against bacteria involved in respiratory disease in cattle: <i>Pasteurella multocida</i>, <i>Mannheimia haemolytica</i>, <i>Histophilus somni</i> (former <i>Haemophilus somni</i>); bacteria involved in acute bovine foot rot (interdigital necrobacillosis); <i>Fusobacterium necrophorum</i>, <i>Bacteroides melanogonicus</i> (<i>Porphyromonas asaccharolytica</i>); and bacteria associated with acute post-partum (puerperal) metritis in cattle: <i>Escherichia coli</i>, <i>Arcanobacterium pyogenes</i> and <i>Fusobacterium necrophorum</i>. PHARMACOKINETICS After administration, ceftiofur is quickly metabolised to desfuroylceftiofur, the principal active metabolite. Desfuroylceftiofur has an equivalent anti-microbial activity to ceftiofur against the bacteria involved in respiratory disease in swine. The metabolite is further metabolised by both animals into inactive bodies. Due to transportation in white progenitor cells, the active component is present at the site of infection, is active and remains active in the presence of necrotic tissue and debris. In pigs given a single intramuscular injection of 3 mg/kg body weight, the maximum plasma levels of $3.4 \pm 1.0 \mu\text{g/ml}$ are reached within 1–2.5 hours. No accumulation of desfuroylceftiofur has been observed after daily doses of 3 mg ceftiofur/kg bw intramuscularly, mean plasma concentrations of metabolite were $3.4 \pm 1.0 \mu\text{g/ml}$ at 1 hour, $2.2 \pm 0.7 \mu\text{g/ml}$ at 4 hours and $1.1 \pm 0.4 \mu\text{g/ml}$ at 8 hours. The MIC₉₀ values of ceftiofur against the target swine pathogens range between 0.12–2.0 $\mu\text{g/ml}$. In cattle, after administration of 1 mg ceftiofur/kg bw subcutaneously to the ear, maximum plasma levels of $2.55 \pm 1.11 \mu\text{g/ml}$ are reached within 0.33 hours. The half-life (<i>t</i>_{1/2}) of ceftiofur administered in cattle is 15.5 ± 7 hours. No accumulation of the active metabolite is observed after 5 consecutive daily injections of 1 mg/kg bw. Ceftiofur and its metabolites are rapidly eliminated, mainly via the urine (more than 70%) and to a smaller extent in the faeces. The metabolite binds to serum proteins (>90%) but is rapidly eliminated. After injection, ceftiofur is distributed in the extracellular body fluids and tissues, especially the lungs. INDICATION Pigs: Treatment of bacterial respiratory diseases associated with <i>Pasteurella multocida</i>, <i>Actinobacillus pleuropneumoniae</i> and <i>Streptococcus suis</i>. Cattle: Treatment of bacterial respiratory disease associated with <i>Pasteurella multocida</i>, <i>Mannheimia haemolytica</i> and <i>Histophilus somni</i>. Treatment of interdigital necrobacillosis (bovine foot rot) associated with <i>Fusobacterium necrophorum</i> and <i>Bacteroides melanogonicus</i> (<i>Porphyromonas asaccharolytica</i>). Treatment of acute post-partum (puerperal) metritis within 10 days after calving associated with <i>Escherichia coli</i>, <i>Arcanobacterium pyogenes</i> and <i>Fusobacterium necrophorum</i>, sensitive to ceftiofur, in cattle. The indication is restricted to cases where treatment with another antimicrobial has failed. ROUTE OF ADMINISTRATION Pigs: Intramuscular injection Cattle: subcutaneous injection. 3 RECOMMENDED DOSE SHAKE WELL BEFORE USE. Pigs: 3 mg ceftiofur/kg bw/day for 3 days via intramuscular route, i.e. 1 ml/16 kg bw at each injection. Cattle: - Respiratory disease: 1 mg ceftiofur/kg bw/day for 3 to 5 days by subcutaneous injection, i.e. 1 ml/50 kg bw at each injection. - Acute interdigital necrobacillosis: 1 mg/kg bw/day for 3 days by subcutaneous injection, i.e. 1 ml/50 kg bw at each injection. - Acute post-partum metritis within 10 days after calving: 1 mg/kg bw/day for 5 consecutive days by subcutaneous injection, i.e. 1 ml/50 kg bw at each injection. In case of acute post-partum metritis, additional supportive therapy might be required in some cases. CONTRA-INDICATION Do not use in case of hypersensitivity to ceftiofur or other <i>β</i>-lactam antibiotics. Do not use intravenously. Do not use in poultry due to risk of spread of antimicrobial resistance to humans. INTERACTION WITH OTHER MEDICINAL PRODUCT: None Information PREGNANCY AND LACTATION Even though studies in laboratory animals show no evidence of teratogenesis, abortion or influence on reproduction, the reproductive safety of ceftiofur has not been specifically investigated in pregnant cows or sows. Use only according to a benefit/risk assessment by the responsible veterinarian. SIDE EFFECTS: Hypersensitivity reactions unrelated to dose can occur. Allergic reactions (e.g. skin reactions, anaphylaxis). SYMPTOMS AND TREATMENT OF OVERDOSE Cephalosporin overdoses are unlikely to cause significant problems other than GI distress, but other effects are possible. Use of dosages in excess of those labeled or by unapproved routes of administration may cause violative residues. STORAGE Store in cool dry place at a temperature $\leq 25^{\circ}\text{C}$. Strictly avoid light and heat exposure. WARNING & PRECAUTION Special precautions for use in animals: <i>Shake the bottle well before use</i> to bring the product back into suspension. In case of the occurrence of allergic reaction the treatment should be withdrawn. CEPTIFI selects for resistant strains such as bacteria carrying extended spectrum <i>β</i>-lactamases. These bacteria may pose a risk to human health if they spread to humans, e.g. via food. Whenever possible, CEPTIFI should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly to narrower spectrum antimicrobials. The use of CEPTIFI is not recommended for the prevention of disease or as part of herd health programs. National antimicrobial policies should be taken into account when the product is used. Increasing incidence of resistance against third and fourth generation cephalosporins (including ceftiofur) has been observed in <i>E. coli</i>. To limit further increase of resistance, CEPTIFI should only be used based on susceptibility testing. Do not use as prophylaxis in case of retained placenta. Inappropriate use of the product may increase the prevalence of bacteria resistant to ceftiofur and may decrease the effectiveness of treatment with other cephalosporins due to the potential for cross-resistance. <i>Special precautions to be taken by the person administering the product:</i> Cephalosporins may cause sensitization (allergy) following injection, inhalation, ingestion or skin contact. Allergic reactions to cephalosporins may lead to cross reactions to penicillins and vice versa. Allergic reactions to these substances may occasionally be serious. <i>Do not handle this product if you are sensitized, or if you have been advised not to work with such preparations.</i> Handle this product with great care to avoid exposure, taking all recommended precautions. If you develop symptoms following exposure, such as skin rash, you should seek medical advice and show the doctor this warning. Swelling of the face, lips or eyes or difficulty with breathing are more serious symptoms and require urgent medical attention. - <i>Keep out of reach of children</i> - <i>Jauhkan daripada kanak-kanak</i> SHELF LIFE: 2 years SHELF LIFE AFTER OPENING: 24 hours WITHDRAWAL PERIOD Pigs: Meat and offal: 6 days Cattle: Meat and offal: 6 days. Milk: zero hours. PACKAGING: 100 ml and 10 ml REVISED DATE: Mar 02, 2026 4  5  Manufactured by: Vemedim Veterinary Drugs and Probiotics Co. Ltd. (Vemedim Animal Health) Song Hau Industrial Zone, Dong Phu commune, Chau Thanh district, Hau Giang Province, Vietnam Product registration holder: VEMEDIM SDN BHD 15, Jalan USJ 1/31, 47600 Subang Jaya, Selangor, Darul Ehsan, Malaysia Tel: (603) 90247118 Fax: (603) 90247168

No.	Previous version	New version
1	Green colour	Red colour
2	Ceftiofur 5000mg/100ml	Ceftiofur 50mg/1ml
3	Layout in one page	Layout in two pages
4	Packaging: 100 ml	Packaging: 100 ml and 10 ml
5	Logo & address: Vemedim Corporation	Logo & address: Vemedim Animal Health